Daily Logic Drill #4

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Disproof by Counterexample · Boundary Case Discovery · Derivative & Function Counterexamples · Universal Refutation.

Section A

1. Provide the minimal positive integer counterexample n to: "For all positive integers n, if n is prime, then $n^2 + 2$ is not prime." *(after TMUA 2022 Paper 2 Q3)*

Ans: 3

2. Find the smallest positive integer a for which there exist positive integers b, c such that $a \mid bc$ but $a \nmid b$ and $a \nmid c$. *(after TMUA 2021 Paper 2 Q4)*

Ans: 4

3. Provide a function $f(x)$ with $f'(x) > 0$ for all $x \in \mathbb{R}$ that serves as a counterexample to: "If $f'(x) > 0$ for all x, then $f(x) > 0$ for all x." *(after TMUA 2018 Paper 2 Q6)*

Ans: $f(x) = x^3$

4. Find a real number $x > 0$ that serves as a counterexample to: "$x > \frac{1}{x}$ for all $x > 0$."

Ans: $\frac{1}{2}$

5. Provide a pair of real numbers (x, y) that serves as a counterexample to: "For all $x, y \in \mathbb{R}$, if $x^2 > y^2$, then $x > y$."

Ans: $(-3, 1)$

6. Find the smallest positive integer n for which $2^n - 1$ is composite despite n being prime.

Ans: 11

7. Provide a pair of non-negative real numbers (a, b) that serves as a counterexample to: "$\sqrt{a+b} = \sqrt{a} + \sqrt{b}$."

Ans: $(1, 1)$

8. Find the smallest positive integer n that serves as a counterexample to: "$n^2 + 1$ is always prime."

Ans: 3

9. Provide a differentiable function $f : [1, \infty) \to \mathbb{R}$ with $\lim_{x\to\infty} f'(x) = 0$ that disproves: "If $\lim_{x\to\infty} f'(x) = 0$, then $\lim_{x\to\infty} f(x)$ exists and is finite."

Ans: "$f(x) = \sqrt{x}$"

10. Find the smallest odd composite integer n that is not a prime power.

Ans: 15

Section B

1. A student claims: "If a polynomial $f(x)$ satisfies $f(n) \in \mathbb{Z}$ for all $n \in \mathbb{Z}$, then $f'(n) \in \mathbb{Z}$ for all $n \in \mathbb{Z}$." Provide a quadratic polynomial $f(x)$ that serves as a counterexample. *(after TMUA 2017 Paper 2 Q16)*

Ans: "$f(x) = \frac{x(x-1)}{2}$"

2. Give a pair of irrational numbers a, b whose sum is rational.

Ans: "$a = \sqrt{2}, b = -\sqrt{2}$"

3. Give a pair of irrational numbers a, b whose product is rational.

Ans: "$a = \sqrt{2}, b = \sqrt{2}$"

4. Provide a pair of functions $f, g : [0, \infty) \to \mathbb{R}$ such that $f(x) \geq g(x)$ for all $x \geq 0$, but $f'(0) < g'(0)$. *(after TMUA 2021 Paper 2 Q12)*

Ans: "$f(x) = 10, g(x) = x$"

5. Find the smallest positive integer n that is a counterexample to: "$n^2 + n + 41$ is prime for all positive integers n."

Ans: 40

6. Provide a counterexample of three finite sets A, B, C showing that $A \cup B = A \cup C$ does not imply $B = C$.

Ans: "$A = \{1\}, B = \{1\}, C = \emptyset$"

7. Find the smallest positive integer n for which $3^n + 2$ is composite.

Ans: 5

8. A student conjectures: "For every odd prime p, the integer $2^p - p$ is divisible by either 3 or 5." Find the smallest odd prime p that serves as a counterexample. *(after TMUA 2021 Paper 2 Q10)*

Ans: 7

9. Provide integers a, b and a modulus $m > 1$ showing that $a^2 \equiv b^2 \pmod{m}$ does not imply $a \equiv \pm b \pmod{m}$.

Ans: $m = 8, a = 1, b = 3$

10. Give a counterexample function $f : [0, 1] \to [0, 1]$ showing that a continuous function need not have a UNIQUE fixed point.

Ans: "$f(x) = x$"

Section C

1. Consider the statement: "$(*)$ If a whole number n is 2 more or 4 more than a multiple of 5, then n is prime." How many counterexamples to $(*)$ are there in the range $1 \leq n \leq 50$? *(after TMUA 2019 Paper 2 Q4)*

Ans: 13

2. Which of the following values of n is a counterexample to the claim: "For all positive integers n, if n is prime, then $n^3 - n$ is NOT divisible by 24?" *(after TMUA 2022 Paper 2 Q3)*

Ans: E) Both $n = 5$ and $n = 7$.

3. A student claims: "If $f : \mathbb{R} \to \mathbb{R}$ is twice differentiable, $f(0) = 0$, and $f''(x) > 0$ for all $x \in \mathbb{R}$, then $f(x) \geq 0$ for all $x \geq 0$." Which of the following functions serves as a counterexample? *(after TMUA 2018 Paper 2 Q6)*

Ans: C) $f(x) = x^2 - 3x$.

4. Consider the statement: "For all positive integers a, b, c, if $a \mid bc$, then $a \mid b$ or $a \mid c$." Which of the following triples (a, b, c) is a valid counterexample? *(after TMUA 2021 Paper 2 Q4)*

Ans: B) $(6, 4, 9)$.

5. Which of the following polynomials satisfies $f(n) \in \mathbb{Z}$ for all $n \in \mathbb{Z}$, but $f'(0) \notin \mathbb{Z}$? *(after TMUA 2017 Paper 2 Q16)*

Ans: D) Both B and C.

6. A student conjectures: "If a continuous function $f : [-\pi, \pi] \to \mathbb{R}$ satisfies $\int_{-\pi}^{\pi} f(x)dx = 0$, then $f(x) = 0$ for all $x \in [-\pi, \pi]$." Which of the following is a valid counterexample? *(after TMUA 2019 Paper 2 Q14)*

Ans: D) All of A, B, C.

7. Consider two statements about functions $f, g : [0, \infty) \to \mathbb{R}$ with $f(0) = g(0) = 0$:

I. $f'(x) \geq g'(x)$ for all $x \geq 0 \implies f(x) \geq g(x)$ for all $x \geq 0$.

II. $f(x) \geq g(x)$ for all $x \geq 0 \implies f'(x) \geq g'(x)$ for all $x \geq 0$.

Which of I and II are true? *(after TMUA 2021 Paper 2 Q12)*

Ans: B) I only.

8. Consider the theorem: "An odd prime p can be written as the sum of two integer squares if and only if $p \equiv 1 \pmod 4$." Which of the following would constitute a counterexample to the **converse** of this theorem? *(after TMUA 2018 Paper 2 Q17)*

Ans: A) A prime $p \equiv 1 \pmod 4$ that cannot be written as the sum of two squares.

Section D

1. Consider: I. "$n = 41$ is a counterexample to '$n^2 + n + 41$ is always prime'." II. "$n = 40$ is the smallest counterexample to '$n^2 + n + 41$ is always prime'." III. "$n = 4$ is a counterexample to '$n^2 + n + 41$ is always prime'." Which of I, II, III are true?

Ans: E) I and II only

2. Prove by contradiction: "There do not exist positive integers a, b with $a^2 - b^2 = 1$."

Ans: Proof by contradiction

3. A student claims: "For every positive integer n, at least one of $n, n+2, n+4$ is divisible by 3." Determine, with proof, whether this claim is true.

Ans: True

4. Give a counterexample to: "If p is prime, then $2^p + 1$ is prime."

Ans: 3

5. Statement $(*)$: "For every positive integer $n \geq 2$, there is a prime strictly between n and $2n$." (a) Verify $(*)$ for $n = 2, 3, 4, 5$ by direct computation. (b) Does this verification constitute a proof of $(*)$? Justify your answer.

Ans: (a) $n = 2 : 3$; $n = 3 : 5$; $n = 4 : 5$ or 7; $n = 5 : 7$. (b) No, finite checking does not prove a universal statement.